

DisasterClaw: A georeferenced simulation and agent framework for budgeted UAV disaster inspection

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Abstract

Aerial disaster agents need an environment in which sensing actions can change the available evidence and their task value can be measured. We present DisasterClaw, a georeferenced simulation and agent framework that converts paired disaster imagery into closed-loop single-UAV inspection episodes. The framework combines map-anchored pre- and post-disaster scenes, a fixed-field-of-view observation renderer, simplified vehicle motion, damage perception, language-conditioned task execution, and episode-level instrumentation. We use DisasterClaw to study budgeted reobservation on paired predictions for 3,753 buildings in 44 regions, quantifying both corrections and harmful changes across three camera footprints. At a reobservation fraction of 0.25, calibrated-entropy allocation raises macro-F1 from 0.6118 to 0.6433, compared with 0.6241 for random allocation and 0.6448 for raw entropy. The framework exposes changes caused by the observation and the allocation rule. The current evidence is limited to rendered orthographic imagery, simplified kinematics, an exposed perception backbone, and an offline allocation endpoint; a corrected closed-loop question-answering evaluation remains future work.

Keywords: UAV disaster inspection, Embodied AI, Simulation platform, Active perception, Budgeted reobservation, Vision-language agents

Review status. This manuscript is an author-review draft. Quantitative tables are reconstructed from archived offline records. The previous online question-answering results were withdrawn after target-grounding and evidence-routing defects were found. Author details and the remaining provenance items in Appendix A require confirmation before submission.

1. Introduction

Disaster response creates a demanding setting for aerial embodied intelligence. An inspection agent must interpret a task, place its sensor over a

relevant region, decide whether the current evidence is sufficient, and return a usable assessment under limited observation and motion budgets. Static remote-sensing benchmarks measure recognition on fixed images, but they do not expose the decision of where and when to observe again. General aerial navigation environments provide motion and language tasks, yet they do not necessarily connect a changed view to a disaster-specific answer. A reproducible bridge between these settings is needed to study whether an agent’s sensing actions improve the decision that motivated them.

The bridge is technically nontrivial. Disaster imagery is often available as georeferenced, orthographic products rather than as a continuously rendered three-dimensional world. Apparent altitude changes must therefore preserve the source-resolution ceiling and maintain a common geographic reference. The agent also operates across several units of analysis: buildings are used by the damage model, regions define task scope, actions incur observation cost, and complete episodes are scored by final answers. Treating these units as interchangeable can make a classifier improvement appear to be an agent improvement. Generation variability adds another ambiguity when a vision-language model can change its answer without any environmental change.

We present DisasterClaw, a georeferenced simulation and agent framework for single-UAV disaster inspection. It transforms paired pre- and post-disaster imagery into an interactive observation environment with a map-anchored scene, a configurable camera footprint, a simplified vehicle state, finite movement and inspection actions, and episode logging. A disaster-response agent receives a natural-language task, a rendered view, and structured spatial evidence; it can search, center on evidence, descend to obtain a narrower footprint, answer, or abstain. The same environment supports an operator-facing console and a headless benchmark runner, so qualitative inspection and quantitative experiments use a shared world and action interface.

DisasterClaw is used here to examine budgeted reobservation. A closer view allocates more source pixels to the target region while discarding some surrounding context. Whether this helps depends on the task: a building label, a count, and a spatial relation can require different evidence. We evaluate the mechanism on paired building predictions while retaining the closed-loop question-answering path as a platform capability. Its corrected task-level evaluation is deferred until a newly reviewed question set is run through the repaired target-grounding interface.

The contributions are as follows.

1. We develop a georeferenced aerial disaster-inspection simulation and evaluation framework that combines paired disaster scenes, a camera-

footprint model, simplified UAV motion, a finite action interface, an interactive console, and headless episode instrumentation. Its physical and data limits are stated explicitly.

2. We implement a closed-loop disaster-response agent that connects task language, rendered observations, building-damage evidence, spatial state, movement tools, and Qwen2.5-VL-7B-Instruct answer generation. Planning, representation, and memory modules are treated as system components rather than isolated algorithmic contributions.
3. We use the framework for a controlled building-level study of budgeted reobservation. The protocol separates corrections from harmful changes under a common selection budget and records the probability-quality consequences of each allocation rule.

The present platform is an imagery-based research environment, not a six-degree-of-freedom flight simulator. It uses orthographic source imagery and simplified motion, and it does not model wind, aircraft dynamics, occlusion, collision avoidance, or battery discharge. These boundaries determine the scope of every experimental claim.

2. Related work

2.1. Aerial embodied agents and simulation platforms

Vision-and-language navigation connects natural-language instructions to sequential visual decisions [1]. AerialVLN extends this problem to UAV navigation, while CityNav introduces large-scale real-world aerial navigation data [2, 3]. OpenFly provides a broader aerial vision-language platform, toolchain, and trajectories across multiple heights and scenes [4]. These works establish language-conditioned aerial motion and height-varying observation as active research areas. DisasterClaw addresses a complementary requirement: constructing task-scoped, paired disaster observations and measuring whether an inspection action changes the requested disaster assessment.

Large pretrained models can be composed with perception and navigation modules without treating the entire stack as a newly learned end-to-end policy [5]. CityNavAgent uses hierarchical semantic planning and global memory, spatial text representations support language-model reasoning in aerial navigation, and AerialClaw exposes modular aerial skills and feedback [6, 7, 8]. DisasterClaw adopts this modular view. Semantic maps, planning, memory, and tool execution support the implemented agent; the platform contribution lies in their integration with georeferenced disaster scenes, controllable inspection observations, and a common measurement interface.

CityEQA connects active urban exploration to embodied question answering and evaluates answers after view refinement [9]. Its setting is a direct precedent for active aerial question answering. DisasterClaw differs in task domain, source imagery, action abstraction, and its paired damage endpoint. These differences preclude a direct numerical ranking, but they motivate reporting environment construction and task scoring with equal precision.

2.2. Disaster interpretation from remote-sensing imagery

xBD provides paired satellite images and building-damage annotations for disaster assessment [10]. RescueADI studies autonomous tool use for adaptive interpretation of remote-sensing disaster images [11]. Thus, paired damage recognition and agentic disaster analysis both predate this work. DisasterClaw uses such static records as the substrate of an interactive inspection environment: a geographic window becomes an observation, vehicle state changes the window, and the resulting episode is scored at both building and task levels.

This transformation remains constrained by the data. Cropping and resampling an orthographic product can alter the number of pixels assigned to a target, but it cannot synthesize unrecorded viewpoints or detail below the native source scale. The platform is consequently suited to controlled studies of spatial coverage, observation allocation, and answer formation. Evidence from it cannot be presented as a measurement of real UAV image formation or field performance.

2.3. Active perception, uncertainty, and task value

Active perception treats sensing as an action selected in service of a task [12]. Planning under partial observability further connects information gathering with uncertain states and action outcomes [13]. This perspective motivates the central platform requirement: a new observation should be evaluated by the decision it enables and the cost it incurs.

Entropy reduction is a convenient trigger but is not equivalent to expected task utility. A distribution can become more concentrated around an incorrect class, while a view may help a count or spatial relation without changing the most uncertain building label. DisasterClaw therefore records classification, correction and harm, final question correctness, and executed observation count as separate endpoints.

Temperature scaling addresses probability calibration, while evaluation under dataset shift shows why in-distribution calibration should not be treated as a universal reliability property [14, 15]. Selective prediction measures the trade-off between coverage and error, and uncertainty-aware language-model planning has been studied through help-seeking behavior [16, 17]. We use

these distinctions to separate calibrated damage probabilities, prediction-set sensing triggers, and the VLM’s generated answer. None alone establishes reliability for the composed agent.

3. The DisasterClaw simulation platform

3.1. Scope and system organization

DisasterClaw is an imagery-based simulation and evaluation framework for single-UAV disaster inspection. Its purpose is to turn georeferenced disaster records into closed-loop episodes in which an agent changes its map position or observation footprint, receives a new image, and is scored on a disaster task. The implementation has four connected layers (Fig. 1): a task interface, a disaster-response agent, a simulated geospatial world, and an instrumentation layer. The operator-facing path uses a React/Vite console connected to a Flask/Socket.IO service. Quantitative experiments use a headless runner that calls the same environment and controller modules without rendering the user interface.

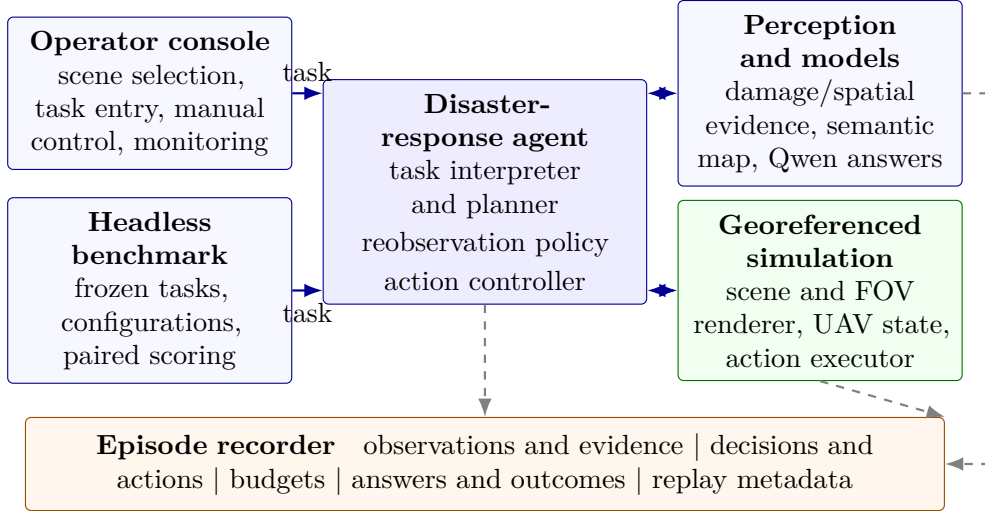


Figure 1: DisasterClaw system organization. The operator console and headless benchmark connect the disaster-response agent to the same perception/model services and georeferenced environment. The recorder retains intermediate model traces, environment transitions, budgets, and task outcomes.

Table 1 states what each layer implements and the boundary attached to it. This distinction is central to the intended use of the platform. Its geospatial state and image renderer support controlled observation studies; they do not reproduce the complete physics of an aircraft or camera.

Table 1: Implemented DisasterClaw capabilities and their evaluation boundaries.

Capability	Implemented function	Boundary in this study
Scene	Georeferenced pre-/post-disaster tiles, building annotations, map anchors, target ROIs, and contextual basemap	Three evaluated events and 44 ROIs; incomplete source archive and no global coverage claim
Observation	Nadir fixed-FOV crop and resampling at three footprint widths; paired archival and current-image channels	Orthographic imagery; no new viewpoint, acquisition time, or detail below the assumed 0.5 m/pixel source scale
Vehicle and actions	Geographic and local position, altitude, heading, speed, task state, straight-line motion, centering, descent, hover, marking, and reporting	One UAV with simplified kinematics; no six-degree-of-freedom dynamics, wind, collision, communication, or battery model
Perception and models	Building localization and four-class damage evidence, spatial summaries, semantic map, planner, and Qwen answer generation	Modular system integration; model revision and some archived runtime fields remain incomplete
Tasks	Operator instructions, damage/presence/count/spatial questions, and archived language-guided navigation instructions	Task-scoped simulation; no claim of unrestricted natural-language or open-world competence
Logging and evaluation	Interactive monitoring, headless episodes, paired task IDs, intermediate evidence, actions, budgets, answers, and terminal scores	New online records are locally represented by supplied summaries; screenshots illustrate function only
Physical boundary	Map-anchored scene transitions and a controlled observation ladder	No photorealistic sensor synthesis, aircraft certification model, hardware-in-the-loop result, or physical-flight validation

3.2. Georeferenced disaster-scene construction

The scene layer reads paired pre- and post-disaster tiles, building polygons, damage labels, and geographic metadata derived from xBD. Pixel-to-geographic and geographic-to-pixel affine transforms place each image on its recorded ground footprint. When a tile becomes active, its center defines the local world anchor. Robot, target, and report coordinates remain available as latitude, longitude, and altitude, while local north and east offsets are recomputed relative to that anchor.

The interactive map can place the active disaster image over a satellite or street basemap, display the tile boundary, and color annotated buildings for operator reference. These annotation overlays are not passed to the evaluated agent as visual input. In rendered observations, georeferenced disaster tiles supply task evidence and the surrounding basemap supplies spatial context when a wider footprint extends beyond the available disaster tile. Reference answers and building-level scores remain restricted to the annotated target ROI, so contextual background is not treated as disaster ground truth.

The evaluated scene population contains three disaster events and 44 target ROIs. This is the experimental subset, not a claim that the software can reproduce every disaster represented by the source dataset. Missing tiles, sparse geographic coverage, affine-registration error, and differences in acquisition date all constrain the resulting environment.

3.3. Observation renderer and camera ladder

The simulated sensor is nadir-facing, with horizontal field of view θ and fixed output width W . For vehicle height h , the ground-footprint width L and effective ground sampling distance g are

$$L(h) = 2h \tan(\theta/2), \quad g(h) = \frac{L(h)}{W}. \quad (1)$$

We use $\theta = 60^\circ$, $W = 1024$, and an assumed source scale of 0.5 m/pixel. A 1024-pixel source tile therefore spans 512 m. Footprints of three, two, and one tile widths define cruise, intermediate, and floor states at approximately 1330.2, 886.8, and 443.4 m, with effective GSDs of 1.5, 1.0, and 0.5 m/pixel (Fig. 2). The heights are derived parameters of this camera abstraction; they are not measured flight altitudes from the source imagery.

For paired damage perception, the post-disaster channel is sampled at the current footprint and fixed output size. The corresponding pre-disaster window is retained at the assumed source scale before the pair is brought to the detector input geometry. This represents access to an archival reference and a controllable current observation. It also means the two channels

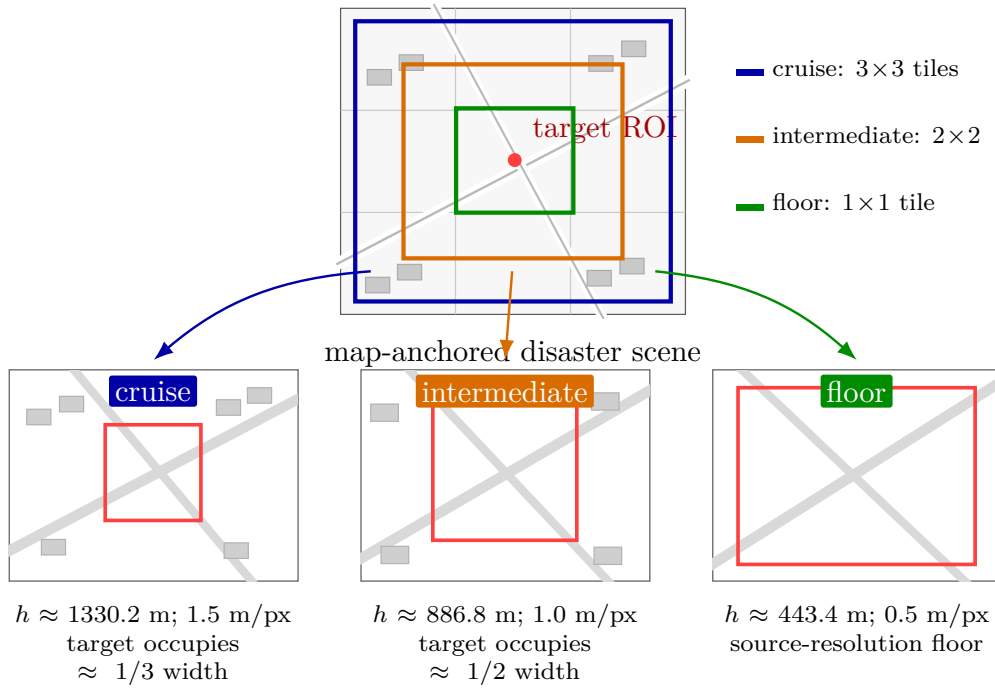


Figure 2: Observation construction from one georeferenced scene. Each state is rendered to 1024×1024 pixels. Descending narrows the ground footprint and assigns more existing source pixels to the target ROI; the floor state reaches, but cannot exceed, the assumed source resolution.

undergo different resampling. The platform records this choice because it can contribute to any observed view response.

Changing height only crops and resamples existing orthographic imagery. The renderer does not create sub-source detail, oblique views, three-dimensional parallax, self-occlusion, atmospheric effects, motion blur, or a new acquisition time. At the floor, further descent cannot add information under the adopted source model.

3.4. Vehicle state and action execution

The world model stores a single UAV with geographic position, altitude, heading, speed, flight status, and task state. It also stores targets, reports, the active disaster tile, map bounds, and timestamps. Motion is implemented by straight-line interpolation in a locally anchored horizontal frame. A position update is converted between geographic coordinates and local north–east offsets, and altitude is recorded as the vertical state.

The task planner and closed-loop agent share a finite set of environment tools: fly to a geographic coordinate, move by a relative horizontal or vertical offset, hover, run disaster detection, mark a target, and produce a report. The inspection controller additionally uses answer, abstain, and stop as episode decisions. A reobservation maps to one or more movement primitives, such as centering on evidence and descending by one camera-ladder state, followed by a new render and perception pass.

This action model supports repeatable studies of where the sensor is pointed and how often another observation is requested. It omits rigid-body attitude dynamics, acceleration limits, wind, terrain following, collision checking, communications, and battery discharge. Executed action count is therefore the principal online cost in this paper. Any time estimate obtained from configured speeds remains a simulator quantity rather than measured endurance.

3.5. Interactive operation and experiment instrumentation

The console exposes disaster selection, a georeferenced situation map, manual flight controls, language-task submission, perception displays, agent state, and event logs. An operator can activate a disaster tile, select a target location, issue an inspection task, observe the planned or closed-loop actions, and inspect the returned damage evidence. Figure 4 illustrates these functions after the agent loop has been defined. The screenshots support interface description and diagnosis; they do not supply quantitative evidence.

The headless benchmark initializes a question and vehicle state, invokes the same observation and action modules, and writes the evolving answer, evidence, controller decision, movement, budget, and terminal outcome. Frozen

question identifiers allow policies to be compared on paired tasks. The implementation also supports a fixed-view mode that holds the environment state constant while repeating answer generation. This instrumentation can separate environmental change from model-generation change in a future corrected evaluation.

4. Disaster-response agent and reobservation mechanism

4.1. Task modes and closed-loop state

DisasterClaw exposes two task modes over the same environment tools. In plan-and-execute mode, a language planner converts an operator instruction into a finite sequence of typed actions with parameters and reasons; a rule-based fallback handles simple geographic tasks when the planner model is unavailable. In closed-loop inspection mode, the agent alternates observation, evidence extraction, decision, and execution until it answers, abstains, stops, or reaches an episode limit. The experiments in this paper evaluate the latter mode.

An inspection task contains a question q , a target region of interest (ROI), and a reference answer y_q . At step t , the environment state $s_t = (x_t, y_t, h_t)$ identifies the UAV’s map position and height. The renderer returns image $I_t = \mathcal{R}(\mathcal{M}, s_t)$ from georeferenced scene $\mathcal{M}$. The agent observation $o_t = (I_t, z_t)$ also includes structured evidence z_t , such as target coordinates, detected-building counts, damage probabilities, and a semantic-map summary. The state notation describes the implemented observation process rather than a complete aircraft dynamics model.

The agent-level decision set is

$$\mathcal{A}_q = \{\text{search, center, descend, answer, abstain, stop}\}. \quad (2)$$

Search and center become relative horizontal movements; descend changes to the next state of the camera ladder. Each movement is executed by the platform before a new image and structured state are returned. The task score is assigned only to the terminal answer, while the episode log retains intermediate answers, decisions, and costs.

Figure 3 shows the resulting state machine. The evidence-sufficiency test and the budget/action-feasibility test separate a model decision from an executable environment transition. The same interface can repeat answer generation while holding the complete environment observation fixed.

4.2. Perception, spatial evidence, and answer generation

The evaluated damage backend is recorded as `xview2_first`, with a single ResNet-34 configuration and seed 0. It processes pre-/post-disaster

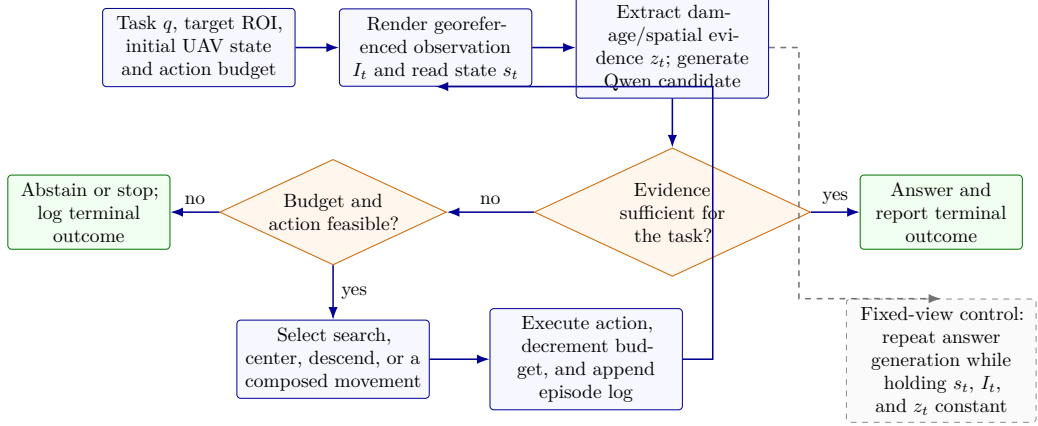


Figure 3: Closed-loop disaster-response agent and reobservation state machine. A reobservation is counted only after a feasible movement is executed and a new georeferenced observation is rendered. The dashed branch denotes the available unchanged-view diagnostic.

image pairs and supplies detected building locations and four-class damage probabilities: no damage, minor damage, major damage, and destroyed. The backend is shared within the principal offline comparisons. Its prior exposure to evaluation events is disclosed in Section 5.

Predictions inside the target ROI are matched to annotated buildings by a greedy, one-to-one nearest-centroid rule with a 12 m distance threshold. An unmatched building receives a no-damage probability vector before normalization and temperature scaling. Consequently, the reported building endpoint combines localization, matching, and damage classification. We retain this archived convention, disclose unmatched counts, and add a matched-only sensitivity analysis.

For online inspection, the agent receives the current image, target scope, structured counts, damage evidence, and a semantic-map snapshot. Spatial representations, hierarchical planning, and memory can support grounding and search [6, 7, 5]; in this study they are system components whose individual navigation effects are not isolated. The final structured question-answering call uses the locally loaded Qwen2.5-VL-7B-Instruct model. Its output is normalized against the task’s answer vocabulary before scoring.

This modular organization keeps the decision boundary visible. A perception module can change a building label without repairing the requested answer, and Qwen can vary its answer without a new environment observation. The episode record preserves both stages so those cases can be distinguished.

4.3. Uncertainty-guided reobservation

Let r_{ij} denote an uncalibrated probability for damage class j . Temperature scaling produces

$$p_{ij}(T) = \frac{r_{ij}^{1/T}}{\sum_{\ell=1}^4 r_{i\ell}^{1/T}}, \quad H(p_i) = -\frac{1}{\log 4} \sum_{j=1}^4 p_{ij} \log p_{ij}. \quad (3)$$

The implementation normalizes probabilities, clips small values for numerical stability, and rounds normalized entropy to three decimal places for ranking. A positive scalar temperature preserves the highest-probability class but can change uncertainty ordering between buildings.

The online controller evaluates whether to reobserve after producing a candidate answer and associated perception evidence. A trigger can use raw or calibrated entropy, a prediction set containing more than one class, a fixed allocation, or an expected-entropy lookup. For the expected-entropy rule, the controller computes

$$\tilde{G}_t = \max\{0, H_t - \hat{\mu}(g_{t+1}, \hat{c}_t)\}, \quad (4)$$

and requests reobservation when $\tilde{G}_t > 0.02$, subject to the available budget and action checks. Here $\hat{\mu}$ is a development-set mean entropy indexed by future GSD and predicted damage class. This is a plug-in uncertainty-reduction heuristic, not an estimate of expected question utility given the full task history. Its local damage signal can be misaligned with the evidence needed for a count or spatial question.

An accepted request invokes centering, descent, or both. The next rendered view passes through perception and answer generation, and the controller records the new answer, uncertainty, action, and remaining budget. Valid answers are not rejected solely because their self-reported confidence lies below an answer-confidence threshold; the implemented agent is therefore not described as calibrated selective prediction.

4.4. Offline allocation rules and paired endpoints

The building-level study isolates observation allocation from route construction. Each of N buildings has a cruise probability vector p_i^c and a floor vector p_i^f . For budget fraction b , a policy selects S_b with $k = \text{round}(bN)$ buildings, giving

$$p_i^{(b)} = \begin{cases} p_i^f, & i \in S_b, \\ p_i^c, & i \notin S_b. \end{cases} \quad (5)$$

Each selected building counts as one additional observation in this abstraction. The cost does not represent a route visiting all selected buildings.

Random allocation samples k buildings without replacement. Raw and calibrated entropy select the largest corresponding cruise entropies. Expected entropy reduction fits the mean paired difference $H(p^c) - H(p^f)$ on development buildings, conditioned on cruise predicted class and one of five equal-width entropy bins. Empty cells fall back to a class mean and then the global mean. The prediction-set rule ranks by the number of classes required to reach a frozen adaptive prediction set threshold [18]. A label-informed diagnostic ranks correctable cases first and is used only for retrospective mechanism interpretation.

Let $\hat{y}_i^v = \arg \max_j p_{ij}^v$. A correctable case has $\hat{y}_i^c \neq y_i$ and $\hat{y}_i^f = y_i$; a harmful case has $\hat{y}_i^c = y_i$ and $\hat{y}_i^f \neq y_i$. For a fixed selected set,

$$\Delta \text{Acc}(S_b) = \frac{N_{\text{corrected}}(S_b) - N_{\text{harmful}}(S_b)}{N}. \quad (6)$$

This identity applies to the hard-label endpoint, not to macro-F1, probability quality, or language-task utility.

A future online evaluation will count every task in its denominator, including abstentions as unsuccessful answers, and report executed reobservation actions separately. If a_q is the terminal answer, task accuracy is $|Q|^{-1} \sum_q \mathbf{1}[a_q = y_q]$. The unchanged-view diagnostic holds s_t , I_t , and z_t fixed while repeating the answer-generation call.

Screenshot placeholder (a)

Save the operator-console overview as
`figures/console/console_overview.png`

(a) Operator overview: georeferenced scene, UAV state, task and perception panels.

Screenshot placeholder (b)

Save the closed-loop inspection view as
`figures/console/console_agent_loop.png`

(b) Closed-loop inspection: trajectory, evidence, agent decision, budget and log.

Figure 4: DisasterClaw operator console. Panel (a) summarizes scene selection, the georeferenced map, UAV state, task entry, and perception output. Panel (b) exposes the evolving trajectory, agent decision, evidence, remaining budget, and event log during closed-loop inspection. These screenshots document implemented functions and are not used as quantitative evidence.

5. Experimental protocol

5.1. Evaluation populations and task suite

The study uses two non-pooled populations. The building population measures how the platform’s camera ladder changes damage predictions under controlled view replacement. An archived 200-instruction vision-and-language navigation (VLN) suite diagnoses the integration of planning, recheck, and memory components. Keeping building records and navigation instructions separate prevents distinct units of analysis from being combined into one sample size.

The development building population contains 2,921 buildings from 20 ROIs in Hurricane Harvey and the Mexico earthquake. The principal evaluation population contains 3,753 unique buildings from 44 ROIs: 1,303 in Hurricane Michael, 1,448 in the Moore tornado, and 1,002 in Nepal flooding. The events used to fit allocation statistics and those used for evaluation are disjoint. The perception checkpoint itself does not satisfy that separation: the frozen manifest marks the `xview2_first` backbone as having prior exposure to evaluation events. Results are interpreted within this boundary.

The archived VLN suite contains 200 instructions from Hurricane Michael, the Moore tornado, Nepal flooding, the Palu tsunami, and the Pinery bushfire. Four switch configurations are compared in one recorded run with seed 42: a keyword/greedy baseline (B0), hierarchical semantic planning (B1), uncertainty recheck (B2), and topological memory (B3). Strict success requires terminating within 25 m of the instruction target. Semantic success uses the broader criterion of terminating within 25 m of a ground-truth building with the requested damage class, and navigation error (NE) is the terminal distance to the strict target. Because this suite used an earlier observation stack and one run per configuration, it is retained as a component diagnostic rather than evidence of algorithmic improvement.

The archived building results permit local arithmetic reconstruction from predictions, labels, and correctness flags. The VLN aggregate and episodes are present under `runs/benchmarks/paper_cja_v1/x6_main/` and are transcribed here as author-verified archived results without recomputation. The source image collection and exact training data are incomplete here. Accordingly, the building metrics are independently reconstructed, whereas the VLN suite retains its archived evaluation scope.

5.2. Platform states, policies, and cost control

The camera states use the geometry in Eq. (1): cruise, intermediate, and floor effective GSDs are 1.5, 1.0, and 0.5 m/pixel. The offline budgets are $b \in \{0, 0.1, 0.25, 0.5, 1\}$, with $b = 0.25$ as the reference operating point.

Seven allocation curves are reconstructed: no reobservation, random, raw entropy, calibrated entropy, expected entropy reduction, APS set size, and a label-informed diagnostic.

The frozen calibration parameters are $T = 3.5635948726$ and APS threshold $\hat{q} = 0.8064657026$, with nominal $\alpha = 0.1$. Temperature is selected on the development cruise probabilities by minimizing negative log-likelihood over 241 logarithmically spaced candidates in $[0.25, 4]$. The configured online entropy threshold is 0.7 and the expected-reduction threshold is 0.02.

5.3. Model and execution configuration

The archived VLN suite uses a locally loaded Qwen2.5-VL-7B-Instruct model, as confirmed by the author. The current structured answering interface supports a 300-token output limit and configurable evidence level. Exact generation settings for a replacement Agent-VQA experiment must be frozen in its manifest rather than inferred from backend defaults.

The earlier archived runs use seed 42 and two shards; their manifests record Python 3.11.15, NumPy 2.4.4, PyTorch 2.11, and Transformers 5.5.4. The exact Qwen revision, prompt hash, effective environment, and dirty source snapshot remain unavailable.

Author query: Archive the Qwen checkpoint revision, effective generation configuration, exact prompt, source snapshot, and new episode-level files with the final submission package.

5.4. Metrics and statistical scope

For the building endpoint, we report accuracy, unweighted macro-F1 over four classes, multiclass Brier score, negative log-likelihood (NLL), and expected calibration error (ECE) with 15 equal-width confidence bins. Brier score is the mean sum of squared differences from the one-hot label without division by class count. Corrected and harmed counts follow Eq. (6). Metrics use the same building population, including unmatched cases under the archived convention.

We add an exploratory paired ROI bootstrap for two comparisons at $b = 0.25$: calibrated entropy versus random allocation and calibrated versus raw entropy. Each of 2,000 resamples draws ROIs with replacement within each observed event, preserving that event’s ROI count. Sampled confusion matrices are aggregated before recomputing macro-F1 differences. The seed is 20260905; reported limits are the 2.5th and 97.5th percentiles. Allocation masks, fitted parameters, and the observed event mixture remain fixed. These intervals address sampling variation among ROIs within the three events, rather than training uncertainty, random-policy seeds, or transfer to new disasters.

The archived VLN table reports Wilson intervals for strict success, semantic success rate (semSR), and mean terminal navigation error. The configurations were executed once with one seed; the table is descriptive and is not used for pairwise significance claims.

6. Platform-enabled evaluation and results

6.1. Historical four-class perception diagnostic

The results in this subsection use the archived `xview2_first` four-class pipeline. They are retained only as historical diagnostics and are not part of, or directly comparable to, the frozen binary ChangeOS evidence chain used by the current main experiment.

The platform generates three observations of the same geographic target scope while varying its footprint and effective GSD. The evaluation retains the same 3,753 building identities across 44 ROIs. Table 2 reports the reconstructed final evaluation population. Moving from cruise to floor increases macro-F1 from 0.6118 to 0.6542 and accuracy from 0.7250 to 0.7423; Brier score and NLL also decrease. The observation control therefore produces a measurable response in the implemented perception pipeline.

Table 2: Building-level results at three simulated camera states on the same 3,753 evaluation buildings. Lower Brier score and NLL are better.

View	GSD (m/px)	Acc.	Macro-F1	Brier	NLL
Cruise	1.50	0.7250	0.6118	0.4814	1.3564
Mid	1.00	0.7357	0.6327	0.4669	1.3314
Floor	0.50	0.7423	0.6542	0.4616	1.3073

Across cruise and floor, 383 buildings change predicted class. Of these, 191 change from incorrect to correct and 126 from correct to incorrect; 66 change between incorrect classes. The net accuracy gain is 65/3,753, or 1.73 percentage points. Thus, a narrower footprint does not act as a uniformly beneficial treatment. It creates both correction opportunities and harmful changes that an allocation rule must distinguish.

Localization contributes to this endpoint. The cruise, intermediate, and floor records contain 912, 916, and 907 unmatched buildings. Among the 191 corrected cases, 35 involve an unmatched record in at least one view; among the 126 harmed cases, 18 do so. On the 2,757 buildings matched at all three views, macro-F1 still rises from 0.7069 to 0.7489 (Table A.1). The response therefore persists in the matched-only subset, although that selected population cannot replace the full endpoint.

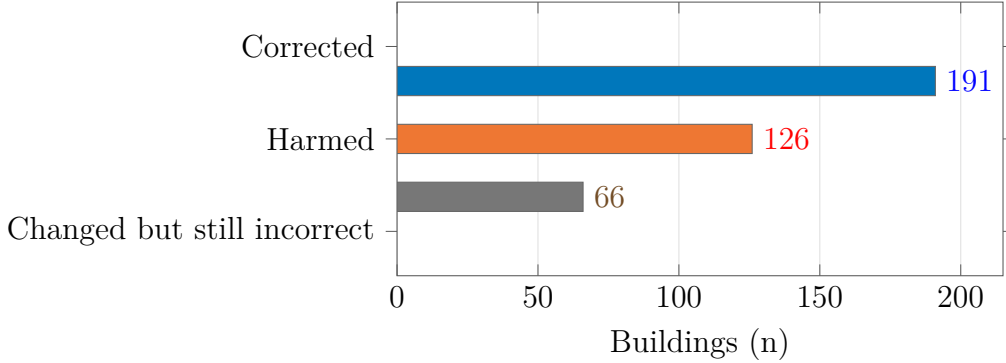


Figure 5: Outcome decomposition for the 383 buildings whose predicted class changes from the cruise to the floor view. The categories are mutually exclusive. Counts are 191 incorrect-to-correct changes, 126 correct-to-incorrect changes, and 66 changes between incorrect classes. The figure describes the recorded imagery-based perception response; it is not a physical-flight treatment effect.

These results validate the platform’s use for a controlled observation-allocation study. They do not validate the assumed altitude against the original sensor, and they do not show that a physical revisit would reproduce the same change.

6.2. Historical four-class allocation diagnostic

At $b = 0.25$, each active rule selects 938 buildings (Table 3). Calibrated entropy obtains macro-F1 0.6433, compared with 0.6241 for random allocation and 0.6118 without reobservation. It produces 53 net additional correct predictions, versus 19 for the recorded random allocation. Uncertainty-based ranking can therefore identify useful reobservations in this observed building population under a common selection budget.

Table 3: Offline allocation at $b = 0.25$. All active rules select 938 floor-view predictions and are scored with the same calibrated vectors. Net correct counts corrections minus harms. The label-informed row uses evaluation labels.

Policy	Actions	Acc.	Macro-F1	ECE	Net correct
No reobservation	0	0.7250	0.6118	0.2423	0
Random	938	0.7301	0.6241	0.2437	19
Raw entropy	938	0.7397	0.6448	0.2578	55
Calibrated entropy	938	0.7391	0.6433	0.2609	53
Expected entropy reduction	938	0.7357	0.6308	0.2551	40
APS set size	938	0.7343	0.6367	0.2549	35
Label-informed diagnostic	938	0.7759	0.6928	0.2853	191

Raw entropy is numerically strongest among deployable rules at this budget, with macro-F1 0.6448. Expected entropy reduction reaches 0.6308 and APS set size reaches 0.6367. The experiment therefore does not establish an advantage for calibrating the ranking score or for replacing uncertainty with the fitted entropy-reduction proxy. The label-informed diagnostic performs better at intermediate budgets by selecting corrections and postponing harmful cases, but it uses evaluation labels and is not deployable.

Figure 6 places the reference operating point in the complete recorded budget sweep. The relative ordering is budget dependent, and all policies converge to the same all-floor endpoint at $b = 1$. The label-informed curve is shown only as a retrospective diagnostic.

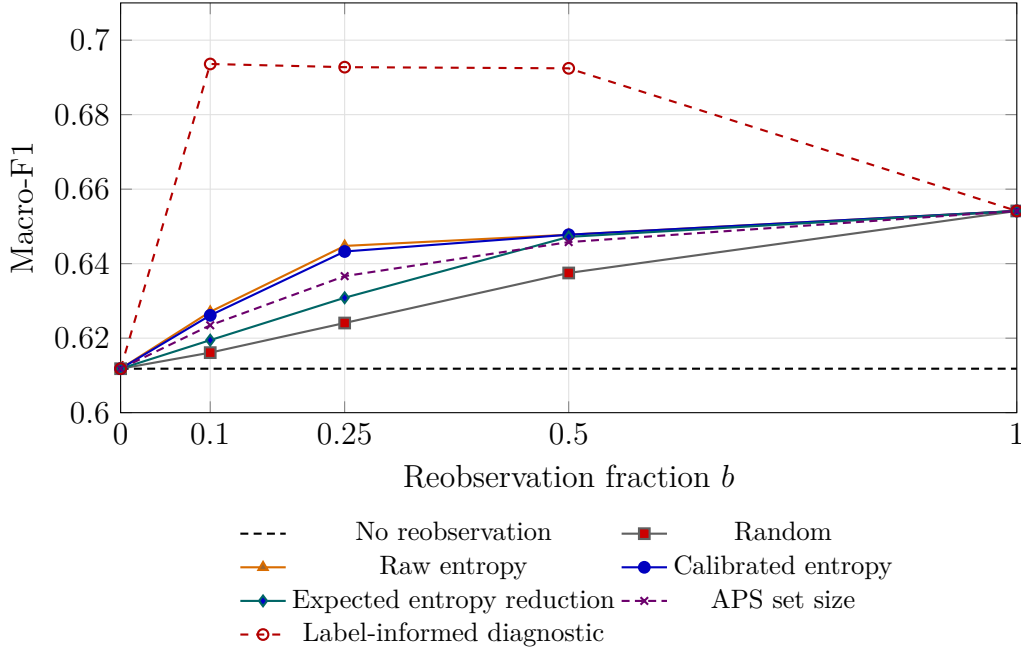


Figure 6: Offline building-level macro-F1 over the recorded reobservation-budget sweep. Each selected building replaces its cruise prediction with its paired floor prediction. The vertical axis is displayed from 0.60 to resolve the observed differences. The random curve is one frozen seeded allocation, the label-informed diagnostic uses evaluation labels, and the action fraction is not a route or mission-cost measure.

The exploratory ROI bootstrap gives a calibrated-entropy minus random macro-F1 difference of 0.0192, with interval $[0.0097, 0.0419]$. The difference from raw entropy is -0.0015 , with interval $[-0.0093, 0.0061]$. These estimates support a conditional distinction from the recorded random selection but no clear benefit over raw entropy. They do not quantify transfer to new disasters.

Probability quality follows a different ordering. At $b = 0.25$, calibrated-entropy allocation raises ECE from 0.2423 to 0.2609 and NLL from 1.3564 to 1.3769, although Brier score decreases from 0.4814 to 0.4763. Improved hard-label performance cannot be described as an overall calibration improvement.

6.3. Archived VLN component diagnostic

The earlier VLN switch experiment provides an additional system-level check (Table A.3). Strict success changes only from 0.070 for the keyword/greedy baseline to 0.080 for hierarchical planning, with overlapping Wilson intervals. Adding uncertainty recheck leaves the reported outcome identical because the recheck mechanism never triggers in B2. Adding topological memory also leaves strict success at 0.080, reduces semantic success from 0.155 to 0.140, and increases mean navigation error from 223.9 to 305.7 m.

This experiment confirms that the planner, recheck, and memory switches were integrated into a complete language-guided navigation path. It does not establish that any of these components improves navigation. In particular, B2 diagnoses an inactive decision path in that archived configuration, and B3 shows that activating memory can alter trajectories without improving strict success. These results support the paper’s emphasis on observable component behavior and evaluation boundaries rather than a claimed VLN performance gain.

6.4. What the platform reveals about reobservation

The retained experiments separate two stages that would otherwise be conflated. First, the camera ladder changes the building-level perception endpoint. Second, budgeted ranking can select a more favorable subset of those changes under a common building budget. Establishing transfer to complete inspection answers requires a new run of the repaired target-grounding and evidence-routing path.

This diagnosis illustrates the value of the framework beyond a single policy score. DisasterClaw makes the environment state, observation transition, perception response, controller action, and terminal answer separately observable. A subsequent task-conditioned policy can therefore be tested on whether it acquires evidence relevant to damage, count, presence, or spatial reasoning, rather than only on whether a generic uncertainty measure decreases.

7. Limitations and deployment boundary

DisasterClaw constructs observations from existing orthographic products. Its camera ladder changes geographic footprint and resampling density, but

it does not synthesize a new viewing angle, time, illumination condition, or detail below the source resolution. Basemap context may come from a different acquisition date, and the pre- and post-disaster channels undergo different resampling. The assumed flat-ground, nadir geometry is unsuitable for claims about oblique imaging, terrain relief, occlusion, or low-altitude structure recovery.

The vehicle model is deliberately lightweight. Straight-line position updates and configured speeds permit reproducible action sequences, but they omit attitude dynamics, wind, inertia, terrain avoidance, collision risk, communication loss, and battery discharge. Executed action count is therefore an experimental cost surrogate. The platform has not been connected to a flight controller or validated against physical UAV trajectories in this study.

The principal perception backbone has prior exposure to the evaluation events. Development/evaluation separation for allocation statistics does not make the complete perception pipeline event-disjoint. In addition, unmatched buildings receive an archived default class, and only three evaluation events are represented. The results characterize the recorded platform configuration rather than generalization to unseen disasters.

The previous closed-loop question-answering runs are excluded because the damage-task target marker was drawn at the image center, target-related detections were filtered out, and the VLM prompt omitted the controller’s target-specific evidence. A replacement study requires the repaired interface, a newly reviewed question set, frozen prompts and model revision, and archived episode-level records. No task-level accuracy claim is made from the withdrawn runs.

Finally, DisasterClaw integrates planning, spatial representation, memory, perception, and language-model components. The archived VLN switch experiment observes these components in an earlier stack, but its single run, inactive recheck branch, overlapping success intervals, and degraded B3 navigation error do not establish component improvements. These modules should be treated as available system capabilities until they are evaluated repeatedly under the current platform configuration.

8. Conclusion

DisasterClaw turns georeferenced pre- and post-disaster imagery into a closed-loop aerial inspection environment. A shared world model connects a fixed-field-of-view renderer, simplified UAV movement, disaster perception, language-conditioned task execution, interactive monitoring, and headless episode evaluation. This organization makes the consequences of a sensing action observable from the perception output through the final task answer.

The budgeted reobservation study demonstrates this evaluation role. Narrower footprints change the building-level endpoint and create both corrections and harms. Entropy-based allocation can select useful changes under a common building budget. The task-level question-answering experiment is withheld until it is repeated with corrected target grounding, evidence routing, and human-reviewed questions.

An archived VLN switch experiment further shows why this traceability matters: enabling planning, recheck, and memory does not by itself establish a navigation gain, and an enabled recheck path can remain inactive. The main result is therefore a platform and measurement framework for studying disaster-inspection agents under controlled observation changes. Its next development should add task-conditioned evidence acquisition, repeated policy runs, complete execution archives, and event-disjoint perception checkpoints before extending the claims to unseen disasters or physical flight.

Declarations for author completion

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CRedit author statement. *To be assigned and approved by the named authors.*

Declaration of competing interest. *To be confirmed by all authors.*

Data and code availability. The working package contains manuscript source and scripts that reconstruct the reported tables from archived predictions and episode logs. Access to the underlying imagery remains subject to its original terms. A public archive, license, anonymization review, and complete data/checkpoint package have not yet been confirmed; a final availability statement must identify the actual release and its restrictions.

Generative AI assistance. Codex was used to assist manuscript restructuring, English drafting, reference checking, and preparation of analysis and typesetting scripts. The present file is an author-review draft. The responsible authors must verify the final content and supply a disclosure consistent with the journal’s current policy; this text does not assert that their verification has already occurred.

Appendix A. Reproducibility and provenance

Appendix A.1. Inputs and reproduction boundary

The manuscript uses a single principal offline population and an archived VLN component diagnostic. The offline source directory is `runs/benchmarks/`

`paper_cja_mech_v1/`. Paired evaluation predictions are read explicitly from `final_fov/fov_ladder_eval_items.jsonl`; development pairs come from `fov_ladder_val_items.jsonl`; the reported allocation curves are in `budget_allocation.json`. This explicit selection prevents a missing evaluation filename from silently causing a fallback to development results.

The previous online Agent-VQA artifacts were removed from the working package after two implementation defects invalidated their use as performance evidence: the displayed damage target was fixed at the image center, and damage evidence filtering rejected every normal damage-class detection. The VLM prompt also omitted target-specific fields already computed by the controller. The removal and rerun prerequisites are recorded in `review/invalidated_online_results.md`; version history remains the recovery source for forensic inspection.

The local scripts `cja_en/scripts/audit_downloaded_results.py` and `cja_en/scripts/build_assets.py` read the original archived inputs without modifying them or invoking a model. They regenerate the offline audit and quantitative assets retained from that audit. The audit records input byte counts and full SHA-256 values. The asset generator rejects changed audited inputs and verifies the allocation reconstruction used by the bootstrap. The system and camera-ladder diagrams are schematics of the implemented framework, not synthetic experimental observations.

The paired-building recomputation matches all seven allocation curves at five budgets for accuracy, macro-F1, ECE, Brier score, NLL, executed selections, and net corrections to an absolute tolerance of 10^{-9} . Final three-view metrics also reproduce to that tolerance. These checks establish numerical consistency of available records. They do not certify that the runtime model, original images, or evaluation history can yet be independently recovered.

Appendix A.2. Software correspondence for the platform description

The platform account in Section 3 is tied to the implementation as follows. Scene state, geographic conversion, xBD indexing, and image assembly are implemented in `backend/world.py`, `backend/geo.py`, `backend/xbd_map.py`, `backend/xbd_store.py`, and `backend/mosaic.py`. The camera abstraction and renderer are implemented in `backend/fov_ladder.py` and `backend/perception.py`. Typed plan execution and the inspection controller are implemented in `backend/ai_planner.py`, `backend/mock_adapter.py`, `backend/agent_vqa.py`, and `backend/recheck.py`; Qwen loading and multimodal calls are contained in `backend/local_qwen_vl.py` and `backend/vlm_analyzer.py`. The operator console is under `frontend/src/`, and the headless protocols used here are under `scripts/benchmarks/`. This map

establishes traceability between the manuscript and the supplied software; it is not a claim that every optional module was active in every experiment.

The present archive does not contain a frozen end-to-end platform release with all imagery, basemap cache, checkpoints, environment variables, and interactive replay artifacts. A complete release should add a scene manifest with affine transforms, a dependency lockfile, an executable episode configuration, and one recorded replay whose observations and actions can be regenerated from hashes.

Appendix A.3. Remaining execution requirements

The original offline frozen manifest was generated on 26 August 2026. Any replacement Agent-VQA run must record the exact model revision, effective environment overrides, prompt hash, committed source snapshot, question-set hash, and human-review state.

The frozen question set and dataset entities are needed to verify their manifest hashes, target construction, scoring rules, and relationships between selection and final populations. Checkpoint files and training manifests are needed to characterize perception exposure directly. Although an additional archive identifies a different event-disjoint perception backend, those results are not combined with the principal tables here: changing the backbone changes the experiment, and its checkpoint provenance remains to be verified.

The replacement run should bind every episode to its source snapshot, question-set hash, human-review record, effective model configuration, prompt hash, and reference result hash, while retaining requested, allocated, and executed budgets separately. The answering mode must be an explicit experiment configuration rather than an inherited default: a raw-image ablation that defaults to structured evidence silently admits perception output into the decision stage. The bench therefore declares the mode per configuration, records it in the run manifest and in every episode record, and reports a deterministic rule baseline separately from the vision-language and hybrid configurations. Aggregation refuses to combine records whose recorded modes disagree.

Appendix A.4. Sensitivity to unmatched detections

Table A.1 holds the population fixed to buildings with a matched detection at every view. Its 2,757 buildings exclude all cases assigned a default vector in any of the three observations. The view response remains positive for accuracy and macro-F1, while Brier score and NLL decrease. This analysis checks one potential source of the overall response; it is not a representative subset selected independently of the evaluated detector. It also does not repair possible centroid-matching errors or perception training exposure.

Table A.1: Sensitivity analysis on 2,757 buildings matched in all three recorded views. This selected subset does not replace the full 3,753-building endpoint.

View	Accuracy	Macro-F1	Brier	NLL
Cruise	0.7918	0.7069	0.3367	0.6455
Mid	0.8052	0.7321	0.3176	0.6123
Floor	0.8099	0.7489	0.3175	0.6103

Appendix A.5. Exploratory uncertainty estimates

Table A.2 reports the added paired ROI bootstrap. It conditions on fixed fitted scores, allocations, and three observed events. ROIs are the sampling units within events; individual buildings are not treated as independent draws. With only three events and prior backbone exposure, these intervals cannot establish robustness to unseen disasters. No multiple-comparison-adjusted confirmatory claim is made.

Table A.2: Exploratory paired ROI-bootstrap macro-F1 differences at $b = 0.25$, with 2,000 resamples and seed 20260905.

Paired comparison	Macro-F1 difference	95% interval
Calibrated entropy – Random	+0.0192	[+0.0097, +0.0419]
Calibrated entropy – Raw entropy	-0.0015	[-0.0093, +0.0061]

Appendix A.6. Archived VLN switch ablation

The VLN diagnostic is stored under `runs/benchmarks/paper_cja_v1/x6_main/`, which contains `results.json`, `episodes.jsonl`, and the B3 memory record. The manifest names the 200-instruction `vln_recheck_eval_v2.json` suite, hybrid grounding, seed 42, and one repeat. The author confirms that its vision-language model was Qwen2.5-VL-7B-Instruct even though the archived environment block leaves the VLM field blank. Table A.3 transcribes the archived aggregate without recomputation. The earlier observation stack and single run prevent causal or cross-seed claims about the enabled modules.

Appendix A.7. Required extensions, without assumed outcomes

TODO(RELEASE): Freeze a runnable DisasterClaw release and publish a scene-to-episode example with a scene manifest, dependency lock, effective configuration, and action/observation hashes. If physical-flight fidelity is claimed later, evaluate the simplified motion and camera abstractions against an external simulator or flight record.

Table A.3: Archived 200-instruction VLN switch ablation. SR is strict success within 25 m of the specified target; semSR accepts a building of the requested damage class within 25 m. Recheck triggers were zero in B0–B3, including the recheck-enabled configurations.

Config.	Enabled navigation components	n	SR (95% CI)	semSR	NE (m)
B0	Keyword/greedy baseline	200	0.070 [0.042, 0.114]	0.165	222.3
B1	Hierarchical semantic planning	200	0.080 [0.050, 0.126]	0.155	223.9
B2	B1 + uncertainty recheck	200	0.080 [0.050, 0.126]	0.155	223.9
B3	B2 + topological memory	200	0.080 [0.050, 0.126]	0.140	305.7

TODO(EXPERIMENT): Generate and independently review a new question set, then repeat the complete moving-policy comparison with the repaired target association and structured-evidence prompt. Archive episode-level records, effective Qwen configuration, prompt hash, source snapshot, and multiple generation seeds.

TODO(EXPERIMENT): Evaluate a backbone whose training and tuning events exclude the final events, with recoverable checkpoints, image/ROI manifests, and an explicit rule for unmatched detections. Report matched-only classification alongside an endpoint that penalizes misses, rather than silently changing the original rule.

TODO(EXPERIMENT): Test task-conditioned evidence selection across damage, count, presence, and spatial questions. Record whether the initial answer was wrong, what evidence was acquired, and whether that evidence addressed the unresolved requirement. Compare at matched executed cost, including movement time and inference cost if measured.

TODO(EXPERIMENT): If operational selective answering is claimed, choose thresholds on separate development data and evaluate risk and coverage on held-out tasks. Retrospective confidence plots in this manuscript do not satisfy that design.

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